USER INTERFACE DESIGN EXP: 10

Create a data visualization (e.g., pie charts, bar graphs) for an inventory management system using javascript

V SAIPRASHANTH 240701457

AIM:

To create data visualizations such as pie charts and bar graphs for an inventory management system using JavaScript and Chart.js.

DESCRIPTION:

This project is an Inventory Dashboard that visually represents stock distribution and item count using:

- **Pie Chart** → shows stock distribution across categories
- **Bar Graph** → shows item count comparison
- **Dynamic Update Feature** → allows user to modify stock values and update charts in real-time

The categories used are:

- Laptops
- Mobiles
- Accessories
- Furniture
- Stationery

PROCEDURE:

Step 1: Create HTML Structure

- Add a dashboard title
- Create two canvas elements:
    - Pie Chart

    - Bar Chart
- Add input fields for updating inventory values
- Add an "Update Charts" button

Step 2: Add Styling (CSS)

- Apply dark theme UI
- Use card layout for charts
- Style input fields and buttons
- Add spacing and alignment for better UI

Step 3: Add Chart.js Library

- Include Chart.js CDN in HTML
- This library is used to generate charts dynamically

Step 4: Implement JavaScript Logic

- Define initial inventory data
- Create:
    - Pie Chart
    - Bar Chart
- Capture user input values
- Update chart data dynamically when button is clicked

ALGORITHM:

1. Start
2. Initialize inventory values
3. Display pie chart using stock distribution
4. Display bar chart using item count
5. Accept updated values from user
6. On button click:
    - Read input values
    - Update chart dataset
    - Refresh charts

7. Stop

PROGRAM:

HTML CODE:

index.html > ...

```html
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>Inventory Dashboard</title>
    <script src="https://cdn.jsdelivr.net/npm/chart.js"></script>
    <link rel="stylesheet" href="style.css">
</head>
<body>
    <div class="container">
        <h1>Inventory Dashboard</h1>

        <div class="charts-grid">
            <div class="chart-card">
            <h2>Stock Distribution</h2>
                <canvas id="pieChart"></canvas>
            </div>
            <div class="chart-card">
            <h2>Items Count Overview</h2>
                <canvas id="barChart"></canvas>
            </div>
        </div>

        <div class="form-card">
            <h2>Update Inventory</h2>
            <form id="inventoryForm">
                <div class="form-group">
                    <label for="laptops">Laptops:</label>
                    <input type="number" id="laptops" value="120" min="1" step="1">
                </div>
                <div class="form-group">
                    <label for="mobiles">Mobiles:</label>
                    <input type="number" id="mobiles" value="300" min="1" step="1">
                </div>
                <div class="form-group">
                    <label for="accessories">Accessories:</label>
                    <input type="number" id="accessories" value="180" min="1" step="1">
                </div>
                <div class="form-group">
                    <label for="furniture">Furniture:</label>
                    <input type="number" id="furniture" value="90" min="1" step="1">
                </div>
                <div class="form-group">
                    <label for="stationery">Stationery:</label>
                    <input type="number" id="stationery" value="220" min="1" step="1">
                </div>
                <button type="button" id="updateBtn">Update Charts</button>
            </form>
        </div>
    </div>
    <script src="script.js"></script>
</body>
</html>
```

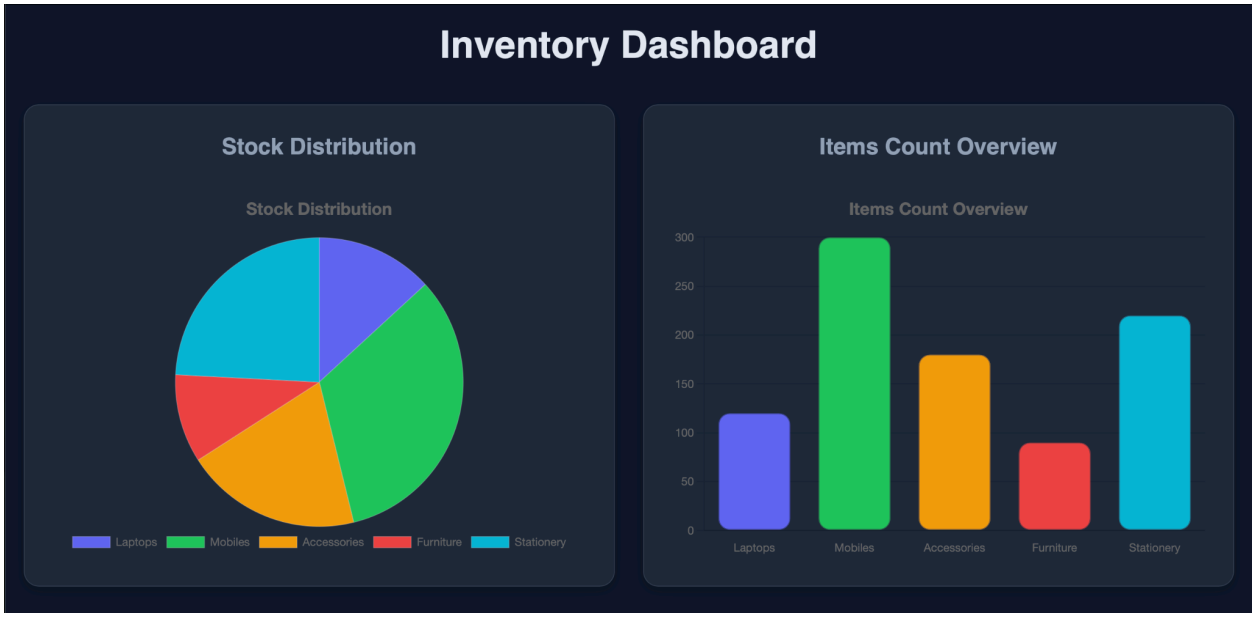
JAVASCRIPT CODE:

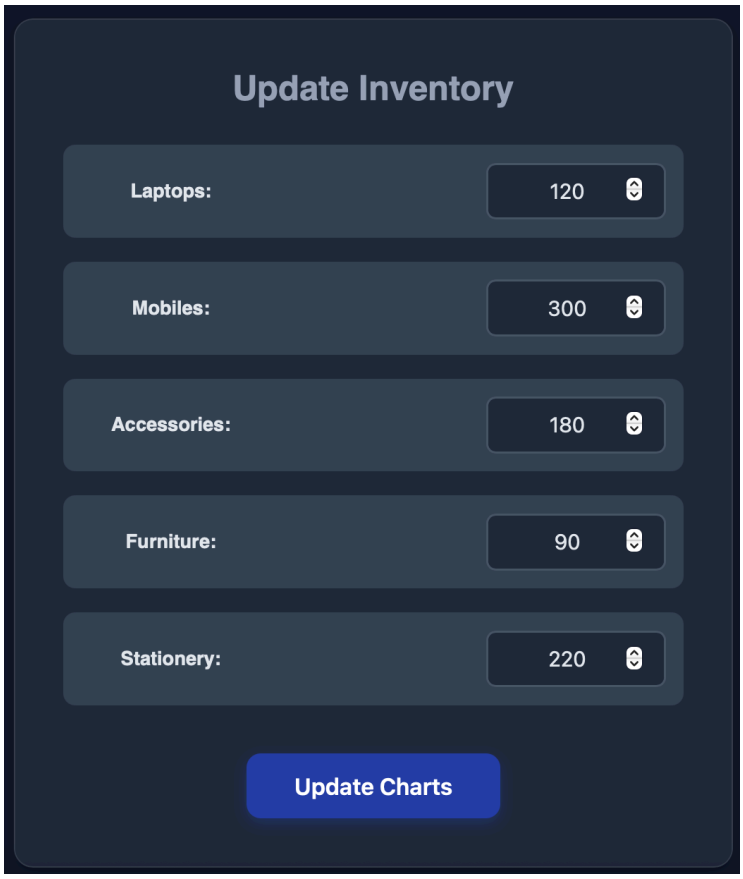
```javascript
const categories = ['Laptops', 'Mobiles', 'Accessories', 'Furniture', 'Stationery'];
let inventoryData = [120, 300, 180, 90, 220];
let colors = ['#6366f1', '#22c55e', '#f59e0b', '#ef4444', '#06b6d4'];

let pieChart, barChart;

document.addEventListener('DOMContentLoaded', function() {
    initCharts();
    setupForm();
});



function initCharts() {
    const pieCtx = document.getElementById('pieChart').getContext('2d');
    pieChart = new Chart(pieCtx, {
        type: 'pie',
        data: {
            labels: categories,
            datasets: [{
                data: inventoryData,
                backgroundColor: colors,
                borderWidth: 2,
                borderWidth: 1,
                borderColor: 'rgba(255,255,255,0.2)'
            }]
        },
        options: {
            responsive: true,
            maintainAspectRatio: false,
            plugins: {
                title: {
                    display: true,
                    text: 'Stock Distribution',
                    font: { size: 18, weight: 'bold' },
                    padding: 20
                },
                legend: {
                    position: 'bottom'
                }
            }
        }
    });

    const barCtx = document.getElementById('barChart').getContext('2d');
    barChart = new Chart(barCtx, {
        type: 'bar',
        data: {
            labels: categories,
            datasets: [{
                label: 'Items in Stock',
                data: inventoryData,
                backgroundColor: colors,
                borderColor: colors.map(c => c + '80'),
                borderWidth: 1,
                borderRadius: 12,
                borderSkipped: false
            }]
        },
        options: {
            responsive: true,
            maintainAspectRatio: false,
            plugins: {
                title: {
                    display: true,
                    text: 'Items Count Overview',
                    font: { size: 18, weight: 'bold' },
                    padding: 20
                },
                legend: {
```

```
                display: false
                }
            },
            scales: {
                y: {
                    beginAtZero: true,
                    grid: {
                        color: 'rgba(0,0,0,0.1)'
                    },
                    ticks: {
                        stepSize: 50
                    }
                },
                x: {
                    grid: {
                        display: false
                    }
                }
            }
        }
    });
}

function setupForm() {
    document.getElementById('updateBtn').addEventListener('click', updateInventory);
}

function updateInventory() {
    const inputs = ['laptops', 'mobiles', 'accessories', 'furniture', 'stationery'];
    let isValid = true;
    const newData = [];

    inputs.forEach((id, index) => {
        const input = document.getElementById(id);
        const value = parseFloat(input.value);

        // Validate positive numbers only
        if (isNaN(value) || value <= 0) {
            input.classList.add('invalid');
            isValid = false;
        } else {
            input.classList.remove('invalid');
            newData.push(value);
        }
    });

    if (!isValid) {
        alert('Please enter valid positive numbers (>0) for all categories.');
        return;
    }

    // Update data
    inventoryData = newData;

    // Update charts with fixed colors
    pieChart.data.datasets[0].data = inventoryData;
    pieChart.update('none');

    barChart.data.datasets[0].data = inventoryData;
    barChart.data.datasets[0].backgroundColor = colors;
    barChart.update('none');

    // Visual feedback
    const btn = document.getElementById('updateBtn');
    const originalText = btn.textContent;
    btn.textContent = 'Updated!';
    btn.style.background = 'linear-gradient(135deg, #28a745, #20c997)';

    setTimeout(() => {
        btn.textContent = originalText;
        btn.style.background = 'linear-gradient(135deg, #007bff, #0056b3)';
    }, 1500);
}
```

OUTPUT:

- Displays:
  - Pie chart for stock distribution
  - Bar chart for item count
- User can:
  - Modify inventory values
  - Click "Update Charts"
  - Charts update dynamically





RESULT:

Thus, the inventory dashboard was successfully created using JavaScript and Chart.js to visualize stock distribution and item counts with dynamic update functionality.